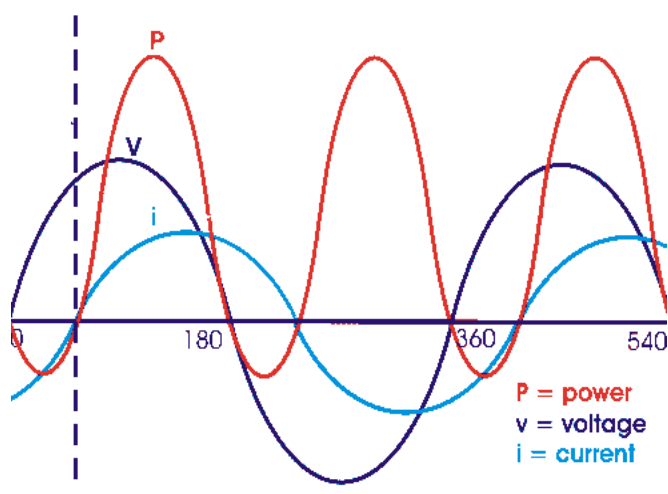




Instruction Manual for Electrical Technology Laboratory **Department of Electrical Engineering, IIT (ISM) Dhanbad**

Experiment No: 05

MEASUREMENT OF POWER IN A SINGLE- PHASE AC CIRCUIT



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21/01/19

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Experiment No.

TITLE: Measurement of power in a single-phase AC circuit.

OBJECTIVE: To measure, directly, the active power consumed by different types of loads using a single-phase wattmeter, find the reactive power, apparent power and power factor of concern load.

APPARATUS USED:

Serial No.	Name of the Equipment	Rating	Model No. of the Equipment	Makers Name
1. 2. 3. 4.				

THEORY:

Wattmeter has two coils, one is called “current coil” and other is “pressure coil”. The current coil carries the load current and pressure coil carries a current proportional to and in phase with supply voltage. The deflection of wattmeter depends upon the currents in the two coils and upon the P.F. of the circuit.

In the wattmeter the current coils are arranged for different ratings i.e. 2.5 A, 5 A, 10 A etc. and voltage coils are rated for 150 V, 300 V, 600 V etc.

While doing the experiment the proper range should be selected according to the load voltage and current. The reading should be multiplied by a factor called the “Multiplying Factor”.

$$\text{Multiplying factor of wattmeter} = \frac{\text{Current range} * \text{Voltage range}}{\text{Full scale reading of wattmeter}}$$

Active Power: - It is the power, which is actually dissipated in the circuit resistance.

$$P = I^2 R = VI \cos \phi \text{ W}$$

Reactive Power: -It is the power developed in the inductive reactance of the circuit.

$$Q = I^2 X_L \sin \phi \text{ VAr}$$

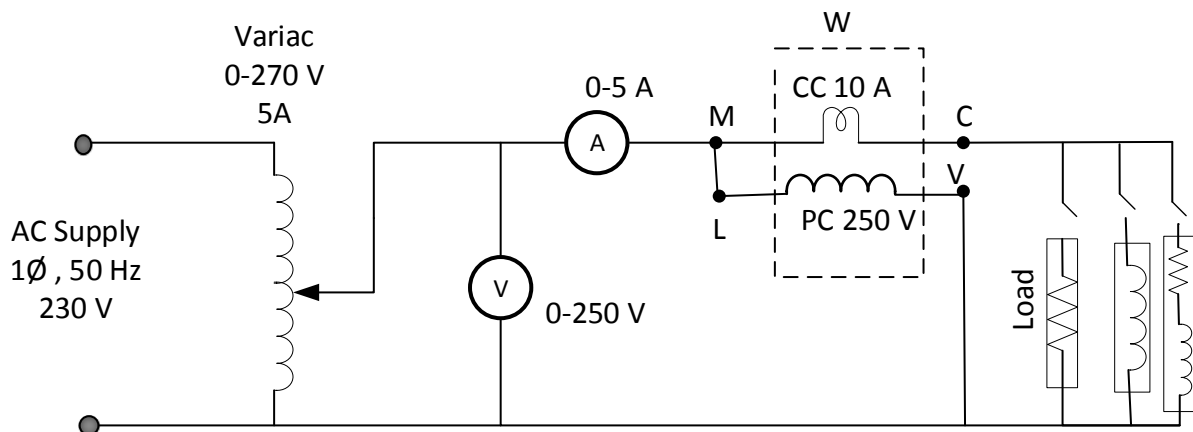
CIRCUIT DIAGRAM:

Fig. 1. Circuit diagram of the experimental setup.

PROCEDURE:**For measuring power in Single-phase AC circuit**

- 1) Choose any load and connect the circuit as shown in Fig.1 make sure the variac is set at 0 volt.
- 2) Switch the power supply and adjust the variac to set a suitable voltage.
- 3) Write down the readings of all meters as shown in the following table consider the proper multiplying factor of the wattmeter- reading as shown on the equipment.
- 4) Take 2 set of readings of different magnitudes of applied voltage to the concern load.
- 5) Complete the experiment for all loads.
- 6) For each set of readings find out the reactive power, apparent power, power factor.
- 7) Draw phasor diagram for each set of readings.

COMMENTS/ DISCUSSIONS:

Write your comments on the results obtained and discuss the discrepancies, if any.

PRECAUTIONS:

- 1) Turn off the power switch to equipment before making connections.
- 2) Don't use broken connecting wires.
- 3) Maintain a work space clear of extraneous material such as books, papers, and clothes.
- 4) Don't wear loose clothes.
- 5) Wear shoes that cover the feet.

REPORT:

- 1) State the difference between Re active power and Active power?
- 2) What is power triangle for an impedance?
- 3) What happens with reactive power for different types of load?
- 4) What is power factor?
- 5) What is apparent power?

DATA SHEET

Experiment No. _____

Date: _____

TITLE: Measurement of power in a single-phase AC circuit.

Name: _____ Sem: _____ Year: _____

Adm. No. _____

Experimental Data: -

Table for single-phase power measurement: -

S.NO.	Load Type	Voltage (V)	Current (A)	Active Power (W)	Power Factor	Reactive Power (VAr)	Apparent Power (VA)
1							
2							
3							
4							
5							
6							

(Signature of the teacher)